Surgical Knowledge Swapping in Large Language Models: Causal Localization and Precision Model Editing

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Abstract

Large Language Models (LLMs) encode factual knowledge within distributed parameter representations learned during large-scale pretraining. However, real-world knowledge evolves continuously, and pretrained models often contain outdated or incorrect information. Traditional fine-tuning approaches modify broad parameter subspaces and may introduce catastrophic forgetting.

In this project, we investigate *surgical knowledge swapping* - the targeted modification of factual associations within pretrained transformer models. We evaluate precision editing algorithms such as ROME and MEMIT on 7B-scale LLMs and systematically measure edit success, paraphrase robustness, locality preservation, logical consistency propagation, and interference under sequential edits. Our goal is to provide a structured evaluation of precision model editing as a scalable alternative to full retraining.

1 Background Information

Large Language Models store factual knowledge implicitly within internal neural representations. Recent interpretability studies (Geva et al., 2020) suggest that transformer feed-forward layers behave like key-value memory stores, where subject representations activate specific value vectors corresponding to factual objects. This conceptualization motivates localized parameter interventions rather than global retraining.

However, maintaining up-to-date factual knowledge in deployed LLMs is challenging. When information changes (e.g., political leadership or scientific updates), retraining is computationally expensive and risks degrading unrelated capabilities. Fine-tuning approaches, including parameter-efficient methods such as LoRA (Hu et al., 2021), still modify broader subspaces and may introduce unintended interference.

Model editing introduces an alternative approach: directly modify the specific parameters responsible for generating a target fact. This surgical approach aims to preserve general capabilities while updating localized knowledge.

2 Literature Review

2.1 Theoretical Foundation

Geva et al. (Geva et al., 2020) demonstrated that transformer MLP layers function as associative memory modules. The first linear layer captures a subject pattern (key), while the second layer maps that representation to a factual value distribution. Knowledge neurons (Dai et al., 2021) further explored how specific neurons encode discrete facts.

2.2 ROME: Rank-One Model Editing

ROME (Meng et al., 2022) introduced a closed-form rank-one update applied to specific MLP weight matrices. Using causal tracing to localize factual recall, ROME replaces subject–relation pairs with new object values while minimizing global parameter disruption. ROME achieves strong paraphrase generalization but may degrade under multiple sequential edits.

2.3 MEMIT: Mass Editing in Transformers

MEMIT (Meng et al., 2023) extends ROME to support large-scale batch editing. By distributing updates across multiple layers, MEMIT improves locality preservation and supports thousands of edits simultaneously.

2.4 Alternative Editing Methods

Other editing approaches include KnowledgeEditor (De Cao et al., 2021) and MEND (Mitchell et al., 2021), which learn gradient transformations to produce localized updates. Surveys of knowledge editing methods (Zhang et al., 2023) categorize these approaches into direct weight editing, meta-learning, and regularization-based methods.

2.5 Knowledge Updating and Unlearning

Recent work such as KnowledgeSmith (Luo et al., 2025) investigates knowledge updating in the context of model editing and unlearning. Surgical Knowledge Rewrite in Compact LLMs (Ngugi, 2025) demonstrates editing robustness in smaller architectures.

Despite these advances, prior work largely evaluates direct edit success. Limited attention has been given to logical consistency propagation and interference under sequential edits. Our project addresses these gaps.

3 Our Approach

3.1 Model Selection

We use a 7B-scale open-weight transformer (Llama-3-8B or Qwen2.5-7B), loaded in 4-bit quantized mode for computational feasibility.

3.2 Dataset

We use the CounterFact dataset introduced alongside ROME (Meng et al., 2022) to perform controlled factual swaps. Additionally, we generate:

- Paraphrase variants via back-translation.
- Logical consistency queries for propagation evaluation.
- Multi-edit scenarios to analyze interference.

3.3 Fact Localization

We apply activation patching and causal tracing (Meng et al., 2022) to identify the MLP layer responsible for factual recall. This ensures that edits target the appropriate internal representation.

3.4 Editing Methods

We implement and compare:

1. ROME for single-fact editing.
2. MEMIT for batch editing.
3. LoRA (Hu et al., 2021) fine-tuning baseline.

3.5 Evaluation Metrics

We evaluate performance using:

- Edit Success Rate (ESR)
- Paraphrase Generalization (PG)
- Locality Leakage (LL)
- Consistency Score (CS)
- Interference Rate (IR)

4 Project Timeline

Weeks 1–2 Literature review and model setup.

Week 3 Implement causal tracing and ROME.

Week 4 Implement MEMIT and batch editing experiments.

Week 5 Implement LoRA baseline and comparative evaluation.

Week 6 Conduct consistency and interference analysis.

Week 7 Final evaluation, visualization, and report writing.

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